DIB isolators, EEPROM, ADC, MCU

File: DIB_EEPROM_ADC_MCU.kicad_sch

DAC, OE, Vref, IOEXP, Temp

File: DAC_Vref_IOEXP_Temp.kicad_sch

Power pre-regulator

File: Power_pre-regulator.kicad_sch

Bias power

File: Bias_power.kicad_sch

Post-regulator

File: Post-regulator.kicad_sch

Rsense protection, power output, Rprog

File: Rsense_protection_power_output.kicad_sch

Mode detection, OVP

File: CC_CV_OVP.kicad_sch

KK1
M3X8/D7985B

KK2
M3X8/D7985B

KK3
M3X8/D7985B

KK4
M3X8/D7985B

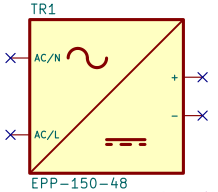
KK5
SK96/84SA

KK6
SP400-0.007-00-104

KK8
B3/BN2842

KK7
M3X8/D7985B

TR1



EPP-150-48

Alternative: Mean Well LOP-200-48

KK9
PCB-MB-01

KK10
PCB-MB-01

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Power module with SMPS pre-regulator, remote sense & programming

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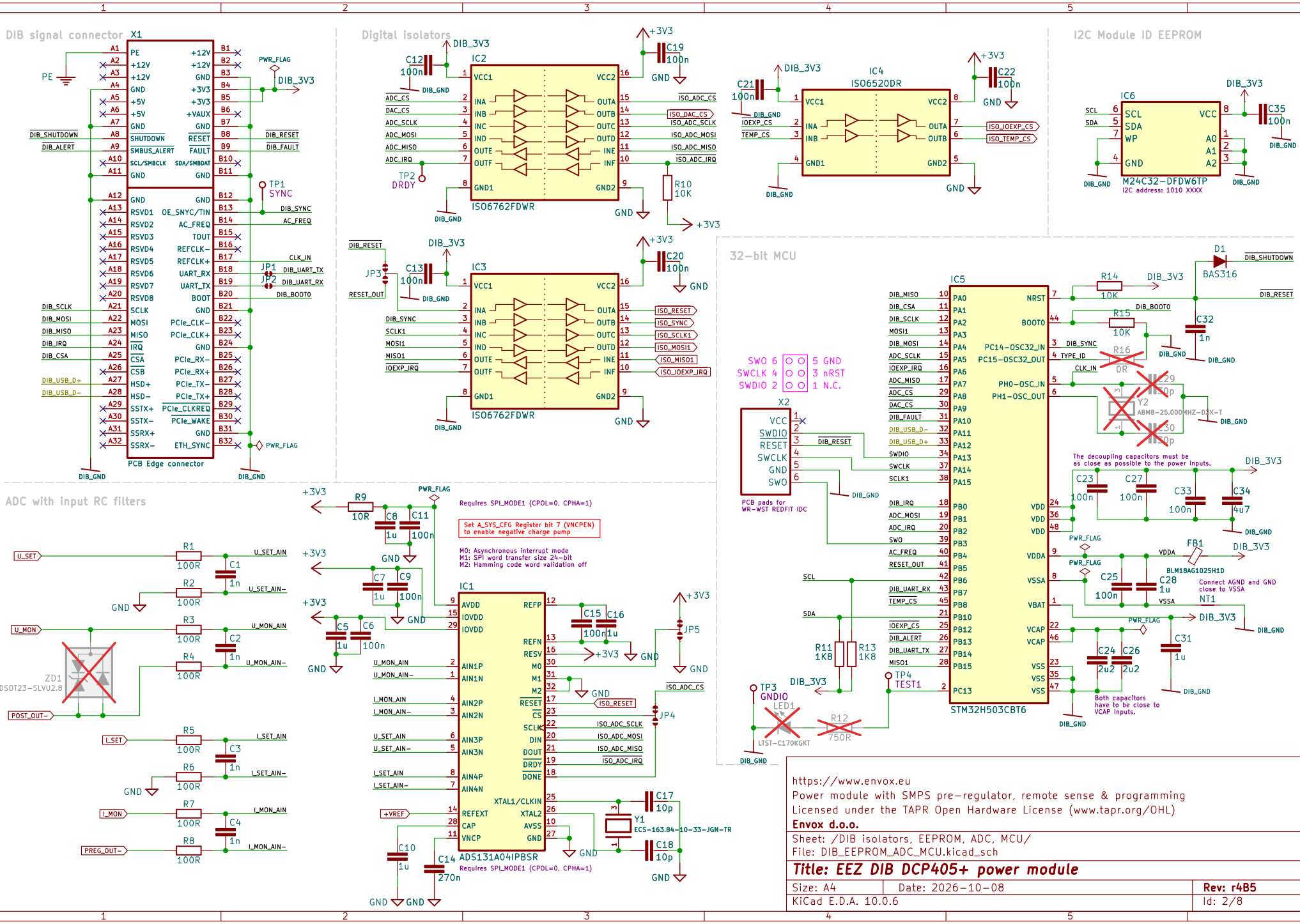
Envox d.o.o.

Sheet: /

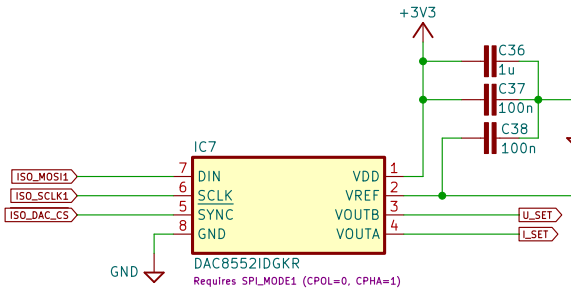
File: EEZ DIB DCP405plus.kicad_sch

Title: EEZ DIB DCP405+ power module

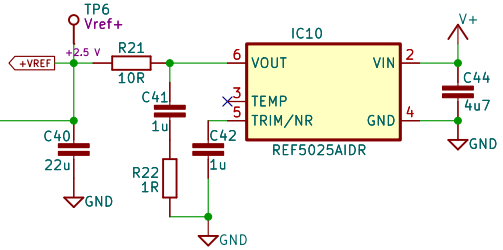
Size: A4	Date: 2026-10-08	**Rev: r4B5**
KiCad E.D.A. 10.0.6		Id: 1/8


DIB signal connector
X1
PCB Edge connector
Digital isolators
IC2
ISO6762FDWR
IC3
ISO6762FDWR
IC4
ISO6520DR
I2C Module ID EEPROM
IC6
M24C32-DFDW6TP
I2C address: 1010 XXXX
32-bit MCU
IC5
STM32H503CBT6
PCB pads for WR-WST REDFIT IDC
ADC with input RC filters
IC1
ADS131A04IPBSR
Requires SPI_MODE1 (CPOL=0, CPHA=1)
Set A_SYS_CFG Register bit 7 (VNCPEN) to enable negative charge pump
M0: Asynchronous interrupt mode
M1: SPI word transfer size 24-bit
M2: Hamming code word validation off
The decoupling capacitors must be as close as possible to the power inputs.
Connect AGND and GND close to VSSA
Both capacitors have to be close to VCAP inputs.
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Sheet: /DIB isolators, EEPROM, ADC, MCU/
File: DIB_EEPROM_ADC_MCU.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-10-08
Rev: r4B5
KiCad E.D.A. 10.0.6
Id: 2/8

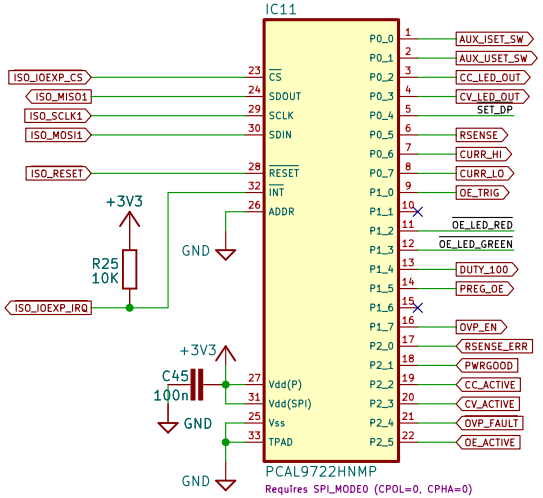
16-bit DAC (set voltage and current)



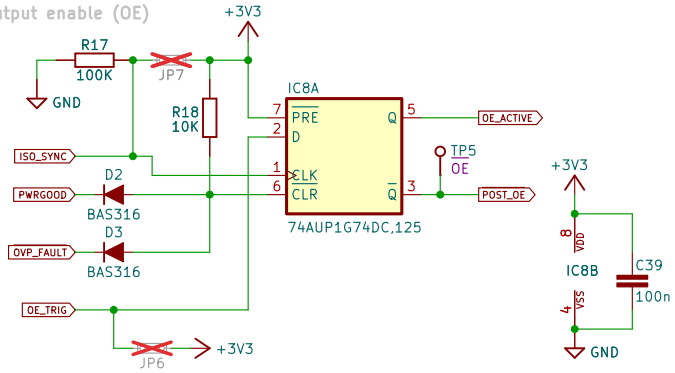
Voltage reference



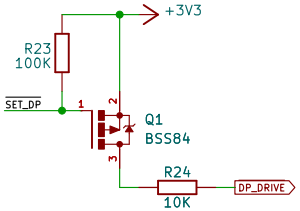
SPI 22-bit I/O Expander



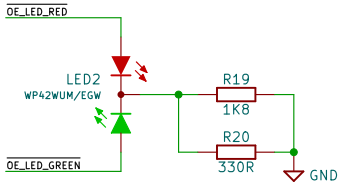
Output enable (OE)



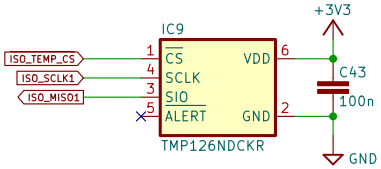
Down-programmer



OE indicator



SPI Temperature sensor

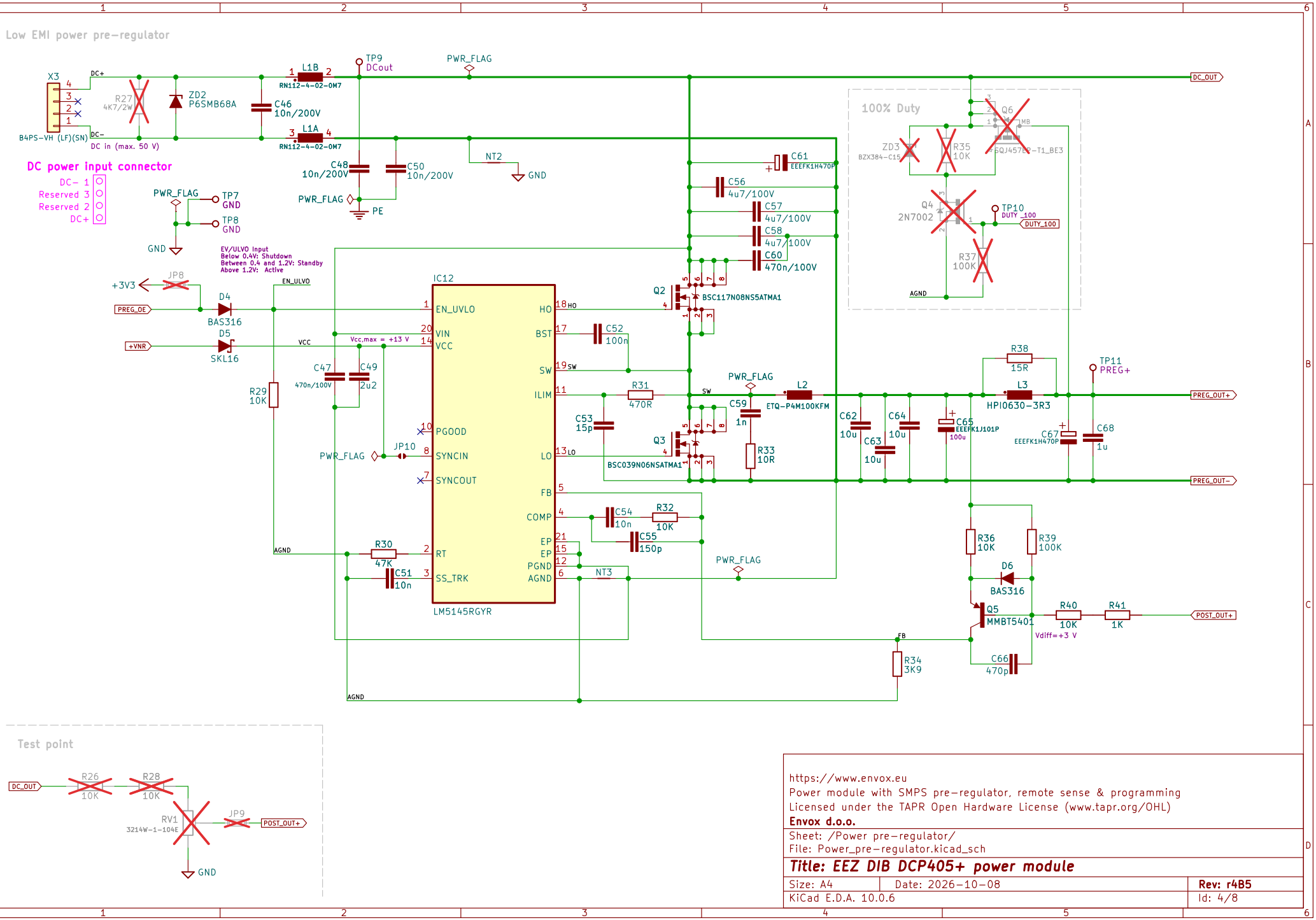


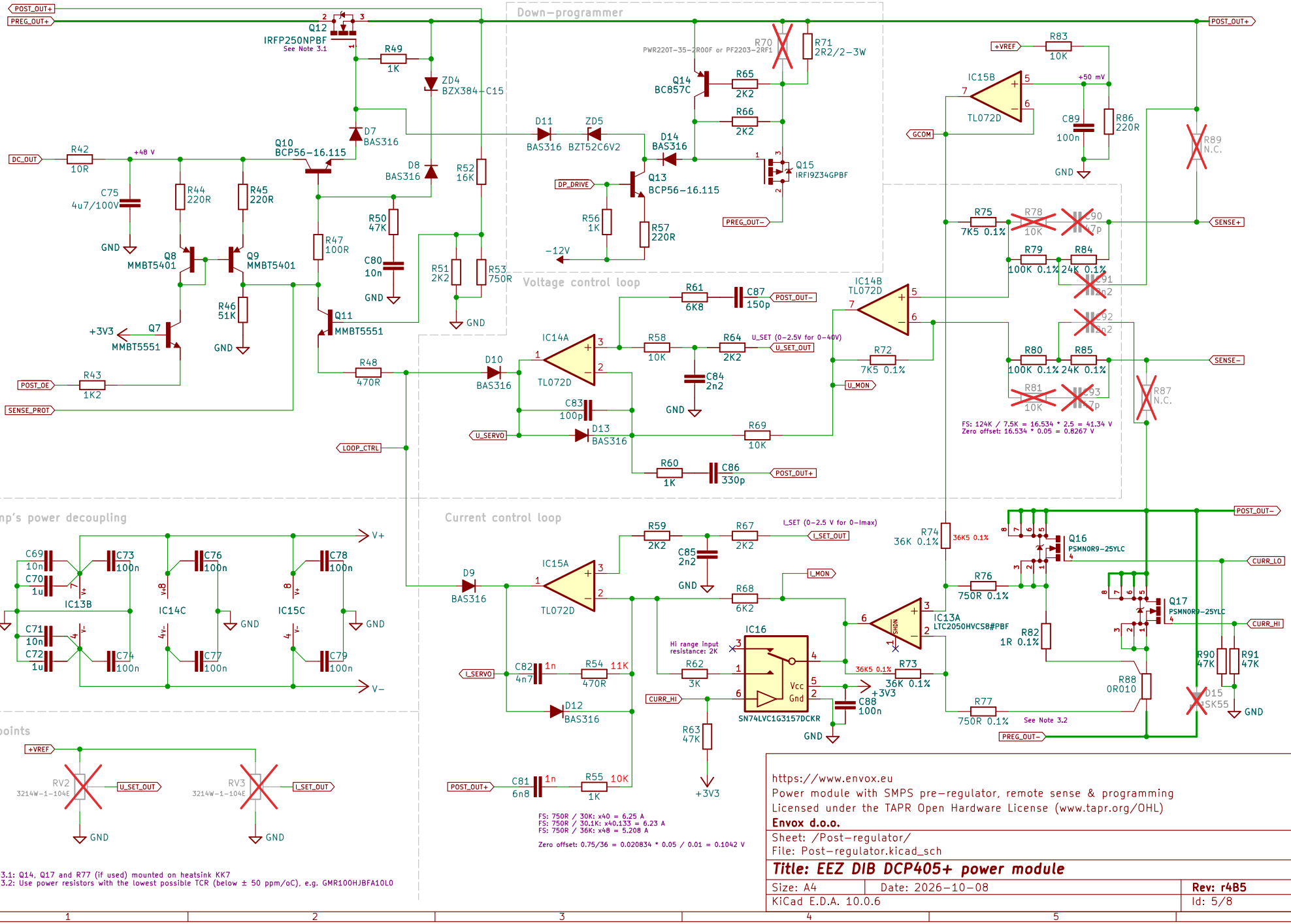
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Sheet: /DAC, OE, Vref, IOEXP, Temp/
File: DAC_Vref_IOEXP_Temp.kicad_sch

Title: EEZ DIB DCP405+ power module

Size: A4	Date: 2026-10-08	**Rev: r4B5**
KiCad E.D.A. 10.0.6		Id: 3/8


Low EMI power pre-regulator
DC power input connector
DC- 1
Reserved 3
Reserved 2
DC+
B4PS-VH (LF)(SN)
DC in (max. 50 V)
X3
R27 4K7/2W
ZD2 P6SMB68A
C46 10n/200V
L1B RN112-4-02-0M7
L1A RN112-4-02-0M7
TP9 DCout
PWR_FLAG
DC_OUT
C48 10n/200V
C50 10n/200V
PWR_FLAG
PE
NT2
GND
C61 EEEFK1H470P
C56 4u7/100V
C57 4u7/100V
C58 4u7/100V
C60 470n/100V
100% Duty
ZD3 BZX384-C15
R35 10K
Q6 SQJ457EP-T1_BE3
Q4 2N7002
TP10 DUTY_100
R37 100K
AGND
PWR_FLAG
TP7 GND
TP8 GND
GND
EV/ULVO Input
Below 0.4V: Shutdown
Between 0.4 and 1.2V: Standby
Above 1.2V: Active
JP8
+3V3
PREG_OE
D4 BAS316
+VNR
D5 SKL16
EN_ULVO
VCC
Vcc,max = +13 V
C47 470n/100V
C49 2u2
R29 10K
IC12
EN_UVLO
VIN
VCC
PGOOD
SYNCIN
SYNCOUT
RT
SS_TRK
HO
BST
SW
ILIM
LO
FB
COMP
EP
EP
PGND
AGND
LM5145RGYR
Q2 BSC117N08NS5ATMA1
C52 100n
R31 470R
C53 15p
Q3 BSC039N06NSATMA1
JP10
C54 10n
R32 10K
C55 150p
R30 47K
C51 10n
NT3
PWR_FLAG
L2 ETQ-P4M100KFM
C59 1n
R33 10R
R38 15R
L3 HPI0630-3R3
TP11 PREG+
C62 10u
C64 10u
C63 10u
C65 EEEFK1J101P 100u
C67 EEEFK1H470P
C68 1u
PREG_OUT+
PREG_OUT-
R36 10K
R39 100K
D6 BAS316
Q5 MMBT5401
R40 10K
R41 1K
POST_OUT+
Vdiff=+3 V
FB
R34 3K9
C66 470p
AGND
Test point
DC_OUT
R26 10K
R28 10K
RV1 3214W-1-104E
JP9
POST_OUT+
GND
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Sheet: /Power pre-regulator/
File: Power_pre-regulator.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-10-08
Rev: r4B5
KiCad E.D.A. 10.0.6
Id: 4/8

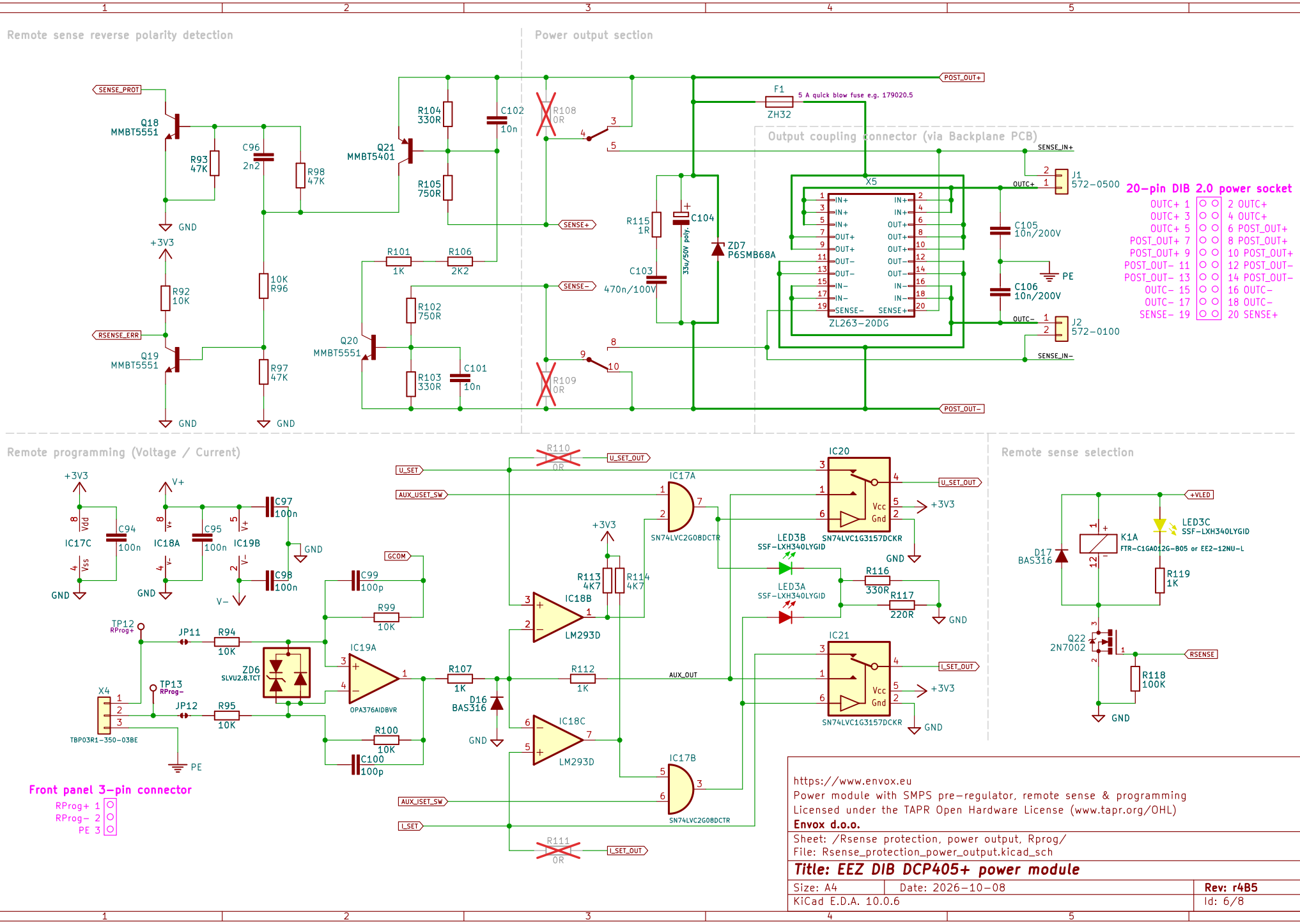


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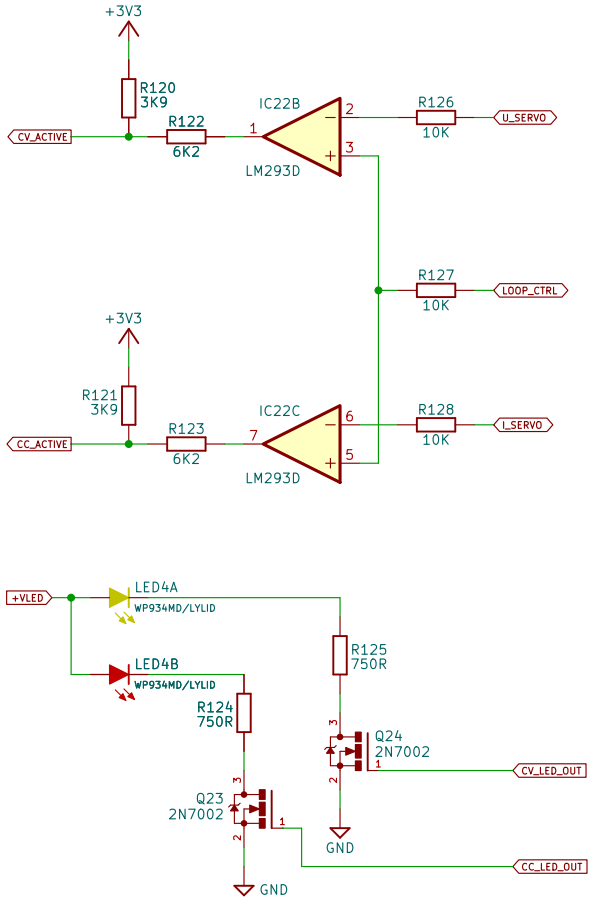
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File: Post-regulator.kicad_sch

Title: EEZ DIB DCP405+ power module

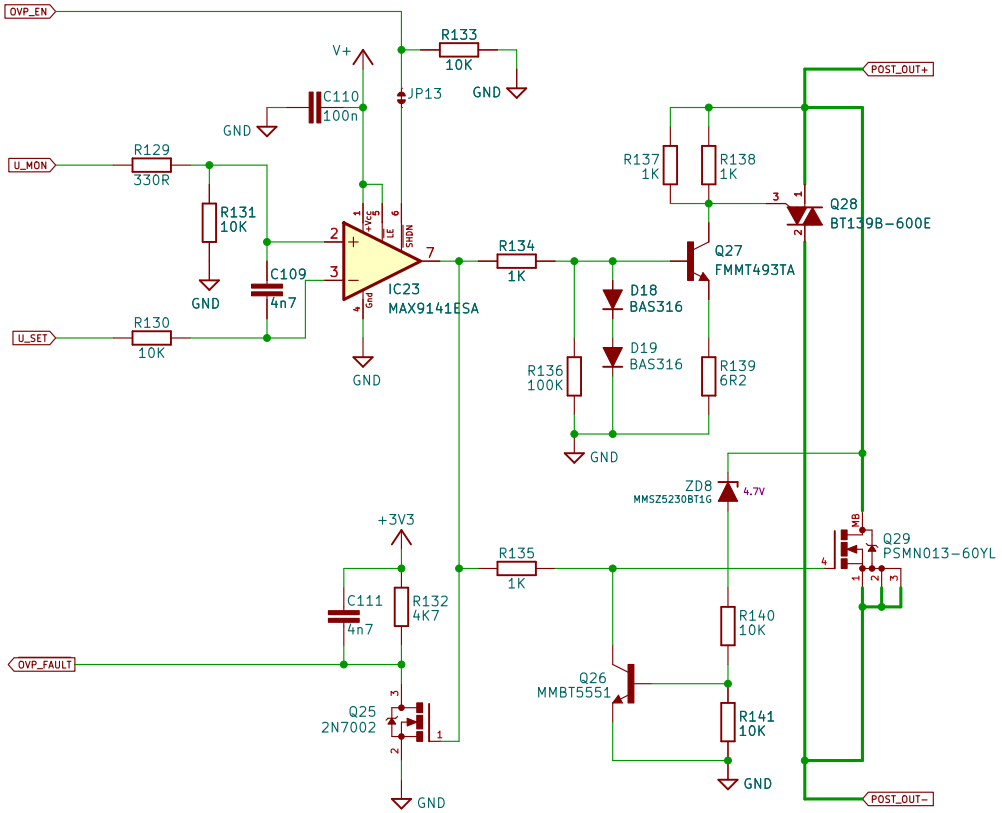
Size: A4	Date: 2026-10-08	**Rev: r4B5**
KiCad E.D.A. 10.0.6		Id: 5/8


Remote sense reverse polarity detection
Power output section
Output coupling connector (via Backplane PCB)
Remote programming (Voltage / Current)
Remote sense selection
SENSE_PROT
Q18 MMBT5551
R93 47K
C96 2n2
R98 47K
Q21 MMBT5401
R104 330R
C102 10n
R105 750R
R108 0R
SENSE+
POST_OUT+
F1
5 A quick blow fuse e.g. 179020.5
ZH32
SENSE_IN+
X5
J1 572-0500
OUTC+
20-pin DIB 2.0 power socket
OUTC+ 1 2 OUTC+
OUTC+ 3 4 OUTC+
OUTC+ 5 6 POST_OUT+
POST_OUT+ 7 8 POST_OUT+
POST_OUT+ 9 10 POST_OUT+
POST_OUT- 11 12 POST_OUT-
POST_OUT- 13 14 POST_OUT-
OUTC- 15 16 OUTC-
OUTC- 17 18 OUTC-
SENSE- 19 20 SENSE+
+3V3
R92 10K
RSENSE_ERR
Q19 MMBT5551
10K R96
R97 47K
GND
R101 1K
R106 2K2
R102 750R
Q20 MMBT5551
R103 330R
C101 10n
SENSE-
R109 0R
R115 1R
C104
C103 470n/100V
33u/50V poly.
ZD7 P6SMB68A
C105 10n/200V
C106 10n/200V
PE
ZL263-20DG
OUTC-
J2 572-0100
SENSE_IN-
POST_OUT-
R110 0R
U_SET
U_SET_OUT
AUX_USET_SW
IC17A
SN74LVC2G08DCTR
IC20
SN74LVC1G3157DCKR
Vcc
Gnd
LED3B SSF-LXH340LYGID
LED3A SSF-LXH340LYGID
R116 330R
R117 220R
IC18B LM293D
R113 4K7
R114 4K7
IC21
AUX_OUT
I_SET_OUT
IC17B
IC18C LM293D
AUX_ISET_SW
I_SET
R111 0R
IC17C
IC18A
IC19B
V+
V-
C94 100n
C95 100n
C97 100n
C98 100n
Vdd
Vss
GCOM
C99 100p
R99 10K
C100 100p
IC19A OPA376AIDBVR
TP12 RProg+
TP13 RProg-
JP11
JP12
R94 10K
R95 10K
ZD6 SLVU2.8.TCT
X4
TBP03R1-350-03BE
Front panel 3-pin connector
RProg+ 1
RProg- 2
PE 3
R107 1K
R112 1K
D16 BAS316
R100 10K
IC19A
OPA376AIDBVR
Remote sense selection
+VLED
LED3C SSF-LXH340LYGID
D17 BAS316
K1A FTR-C1GA012G-B05 or EE2-12NU-L
R119 1K
Q22 2N7002
RSENSE
R118 100K
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Sheet: /Rsense protection, power output, Rprog/
File: Rsense_protection_power_output.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-10-08
Rev: r4B5
KiCad E.D.A. 10.0.6
Id: 6/8

Constant Voltage (CV) and Constant Current (CC) mode detection and indication



Over-voltage protection (OVP)



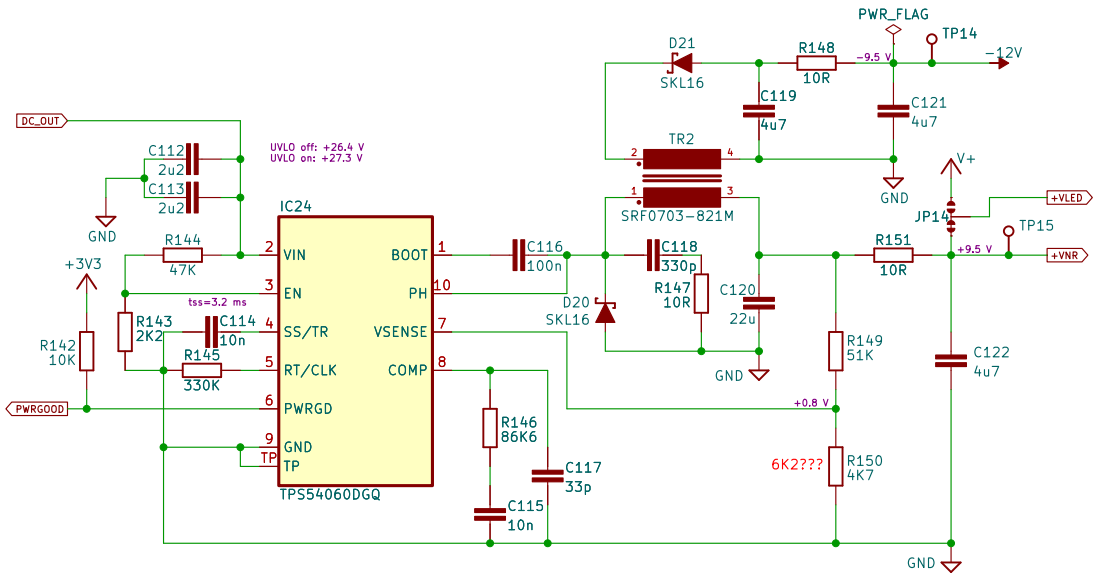
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Sheet: /Mode detection, OVP/
File: CC_CV_OVP.kicad_sch

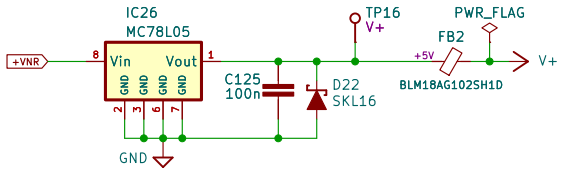
Title: EEZ DIB DCP405+ power module

Size: A4	Date: 2026-10-08	**Rev: r4B5**
KiCad E.D.A. 10.0.6		Id: 7/8

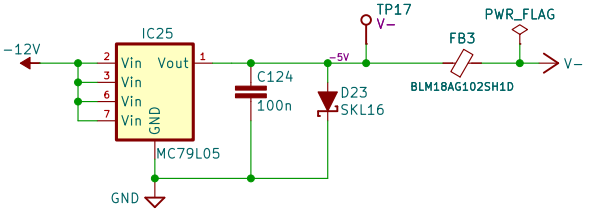
Bias power supply



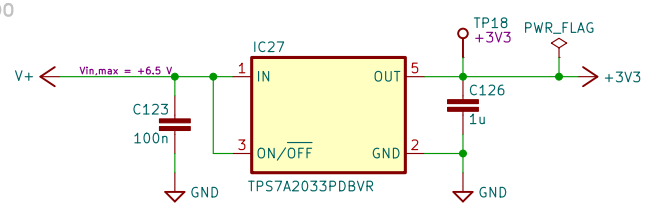
+5V LDO



-5V LDO



+3V3 LDO



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Sheet: /Bias power/
File: Bias_power.kicad_sch

Title: EEZ DIB DCP405+ power module

Size: A4	Date: 2026-10-08	**Rev: r4B5**
KiCad E.D.A. 10.0.6		Id: 8/8